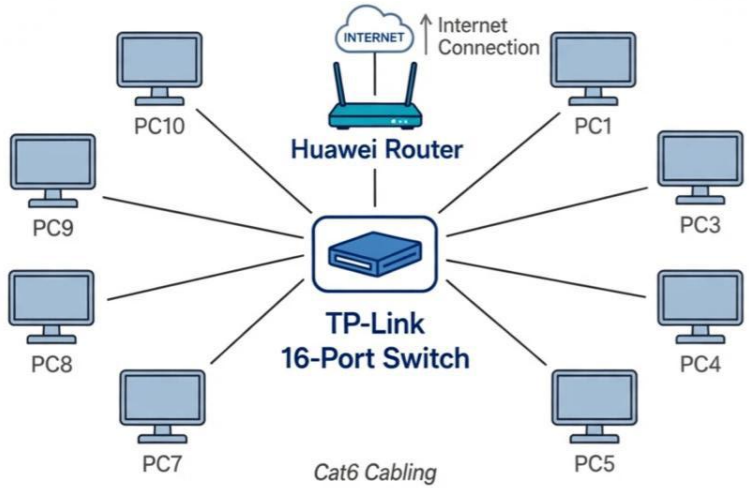
LAN OVERVIEW

LAN Overview - 10 User LAN for Varsity College Computer Lab Room 2

STAR TOPOLOGY DIAGRAM



Star Topology: All 10 PCs are connected to the central switch for centralized communication and sharing

PROJECT
Varsity College Lab

DEVICES
- 10 PCs (Workstations)
- TP-Link 16-port Switch (Central Hub)
- Huawei Router (Internet Gateway)
- Cat6 Cabling (Structured Cabling)

PURPOSE
- File sharing between workstations
- Printer sharing for lab use
- Internet access via router

How to Log In & Operating Procedures

STEP-BY-STEP PROCEDURE

1 UPS
- Switch on UPS

2 Switch
- Switch on Switch (Power LED green)

3 Router
- Switch on Router

4 PC
- Press PC power

Login:
- Username: Studentxx
- Password: Varsity@2026

⚠ **Do not unplug blue cable**

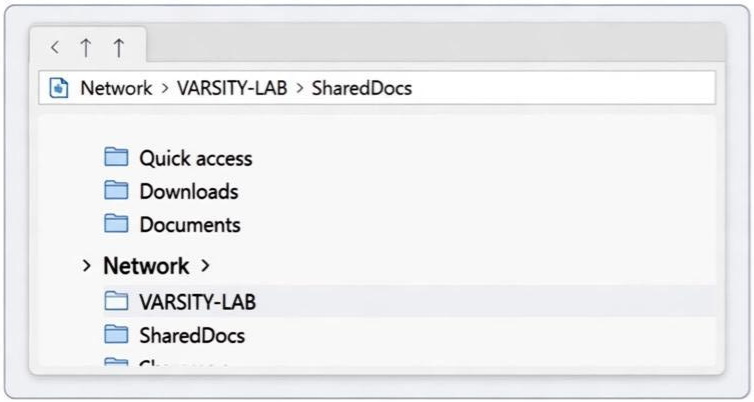
⚠ MANUFACTURER WARNINGS

- ⚠ Only trained personnel are authorized to operate equipment
- ⚠ Ensure all devices are properly grounded before powering on
- ⚠ Keep equipment dry - avoid liquids and moisture near electronics
- ⚠ Use only manufacturer-approved cables and connections
- ⚠ Do not override safety locks or open enclosed units
- ⚠ If any device fails to power on, do not proceed - report immediately

Support Contact:
support@manufacturer.com | +1-800-XXX-XXXX

How to Access Shared Files and Printer

File Explorer — Access Shared Files

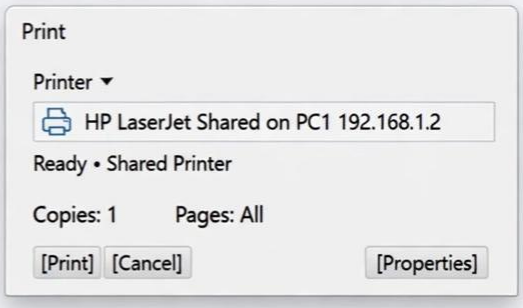


Steps:

1. Open File Explorer and click "Network" in the left vidabll
2. Double-click "VARSITY-LAB" to view available devices/shares
3. Open "SharedDocs" to access shared files

Tip: If prompted for credentials, use your school/domain login.
• IP: 192.168.1.x (shared folder on VARSITY-LAB)

Add & Use Shared Printer



Steps:

1. Open your document and go to File > Print
2. Select printer: "HP LaserJet Shared on PC1 192.168.1.2" from the printer list
3. Click Print to send to the shared printer

Tip: Ensure PC1 is powered on and connected to network 192.168.1.2
• If printer is missing, use "Add Printer" in Settings > Bluetooth & devices > > Printers

Training and Support Options - Justified

1 One-on-one training

Personalized guidance tailored to individual role • Best for initial onboarding • Higher time cost, focused support

2 Group workshop monthly

Scalable for multiple users • Peer learning & Q&A session • Held monthly, 2-hour session, collaborative

3 User manual 2-page

Quick reference guide • Always available offline • Low effort to distribute & update

4 WhatsApp group support

Instant messaging for quick Q&A • Peer & admin responses • Fast, low-cost, accessible on mobile

5 On-site technician

Hands-on support for hardware/setup • Immediate physical troubleshouing • Highest cost, by appointment only

Recommendation: WhatsApp + Manual — recommended (cheapest & fastest)

Covers 80% of common issues • Zero additional budget • Fastest to implement & rollout

User Acceptance, Review and Backup

Review — Requirements Met

Requirement	Status
File sharing	YES
Printer	YES
Internet	YES
Speed 1Gbps	YES

User Signoff

I confirm LAN works — J. Rivera

Signed: 2026-10-04 | Role: End User

Backup & Restore

USB labelted LAN BACKUP 2026-10-04 in cabinet
Securely stored in IT cabinet • Verified 2026-10-04

Router config backup
Config exported and saved to secure NAS location

System restore point
Restore point created — Pre-acceptance snapshot